Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Reading dBc off a Tracking-tab trace.** A cadet walks an HB100 around the PHASER on a 1 m arc with the beam held at broadside and records the Signal vs Time trace. The three lobe peaks read -18.4 dBFS (main lobe), -31.2 dBFS (left first sidelobe), and -29.9 dBFS (right first sidelobe).

   (a) Convert both sidelobe readings to dBc.

   A dBc value is the lobe peak referred to the main-lobe peak, so subtract:

   $$L_{\text{left}} = -31.2 - (-18.4) = -12.8 \text{ dBc}$$
   $$L_{\text{right}} = -29.9 - (-18.4) = -11.5 \text{ dBc}$$

   L_{left} = -12.8 dBc

   L_{right} = -11.5 dBc

   (b) The cadet's partner raises **Rx Gain** by 6 dB and repeats the walk. State what happens to the three dBFS readings and what happens to the two dBc values, and explain why the two answers differ.

   All three dBFS readings rise by about 6 dB as well, because the gain change multiplies every received sample by the same factor. The dBc values do not change, because each one is a difference between two readings taken through the same gain setting and the common term cancels in the subtraction. This is why a pattern is reported as a ratio to its own peak: the result does not depend on the receiver's absolute scale, which is not calibrated here anyway.

   (c) A uniform 8-element array has its first sidelobes at -13 dBc. Comment on the pair measured above: is this array uniformly illuminated, and what accounts for the 1.3 dB spread between the two sides?

   Both readings straddle -13 dBc within about a decibel and a half, which is consistent with uniform illumination measured in a room. A tapered illumination would put the sidelobes at least 17 dB down, and a badly calibrated array would show a much larger imbalance. The 1.3 dB spread between left and right comes from the measurement rather than the array: classroom multipath adds roughly ± 1 dB of ripple, and the walking speed and radius were not identical on the two halves of the arc. The array factor of a symmetric uniform array is symmetric at broadside, so the asymmetry has no home in the physics of the array itself.

2. **Why the time axis is not an angle axis.** The same trace runs from $t = 0.00$ s, when the source was at $-90°$, to $t = 7.40$ s, when it reached $+90°$. The main-lobe peak lands at $t = 3.10$ s. The beam was commanded to broadside.

   (a) Explain why the lobe *amplitudes* on this trace are usable while the lobe *angles* are not.

   The amplitude recorded at any instant is the pattern evaluated at whatever angle the source occupied at that instant, and that value does not depend on how fast the source was moving. Walking slowly through a lobe spends more samples on it and draws it wider on the plot, but the height each sample reaches is unchanged. The angle, in contrast, is never recorded at all. The x-axis stores elapsed time, and recovering angle from time requires knowing the walking speed at every instant, which nothing in the measurement captures.

   (b) Assume a constant walking speed and convert the main-lobe time to an angle. Compare it against the commanded beam angle and state the error.

   A constant speed maps time linearly onto the $180°$ swept:

$$\theta = -90° + 180° \left(\frac{3.10}{7.40}\right)$$
$$= -90° + 75.4° = -14.6°$$

   The beam was commanded to $0°$, so the linear map is in error by $14.6°$.

   θ = $-14.6°$

   error = $14.6°$

   (c) The true half-power beamwidth is $13.1°$. If the hand moves as much as 30% faster or slower than its average while crossing the main lobe, give the range of beamwidths a constant-speed conversion would report.

   A hand moving 30% fast spends $1/1.30$ of the expected time in the lobe, and a constant-speed map converts that shortened interval into a narrower angle. A hand moving 30% slow does the opposite:

$$\theta_{\text{fast}} = \frac{13.1°}{1.30} = 10.1°$$
$$\theta_{\text{slow}} = \frac{13.1°}{0.70} = 18.7°$$

   The reported beamwidth spans nearly a factor of two on a pattern that never changed.

   range = $10.1°$ to $18.7°$

   (d) Name the piece of range hardware that removes this error and state what it records that the hand walk does not.

   A turntable with a shaft encoder. The turntable carries the antenna through a commanded rotation, and the encoder tags every recorded sample with the angular position it was taken at, typically to a small fraction of a degree. The measurement then stores angle directly instead of inferring it from time, so no assumption about rotation speed is needed and a non-uniform rotation costs nothing.

3. **Scan loss and sidelobe asymmetry.** The projected-aperture rule puts the peak power gain of a steered array at $\cos\theta_0$ relative to broadside.

   (a) Compute the predicted peak drop at steer angles of $30°$, $45°$, and $60°$.

   The rule applies to power, so the drop in dB is $10\log_{10}(\cos\theta_0)$:

   $$30° : \ 10\log_{10}(0.866) = -0.6 \text{ dB}$$
   $$45° : \ 10\log_{10}(0.707) = -1.5 \text{ dB}$$
   $$60° : \ 10\log_{10}(0.500) = -3.0 \text{ dB}$$

   $30°$ = -0.6 dB

   $45°$ = -1.5 dB

   $60°$ = -3.0 dB

   (b) A broadside walk peaks at -18.4 dBFS and a $30°$ walk peaks at -19.3 dBFS. Give the measured scan loss, and state whether one pair of walks resolves it given ± 0.5 dB of run-to-run scatter.

   $$\Delta = -19.3 - (-18.4) = -0.9 \text{ dB}$$

   Δ = -0.9 dB

   The measured 0.9 dB sits 0.3 dB from the predicted value of 0.6 dB and well inside the ± 0.5 dB scatter of a hand walk. One pair of runs is therefore consistent with the prediction but does not resolve it: the same pair of numbers would also be consistent with a scan loss anywhere between about 0.4 and 1.4 dB of scan loss. Averaging several walks at each steer angle narrows the scatter enough to make the comparison worth reporting.

   (c) The measured $30°$ trace shows first sidelobes of -12.2 dBc on the low-angle side and -15.2 dBc on the high-angle side. Explain why they are unequal, given that the array factor alone would make them equal.

   The array factor of a uniformly excited array is a function of $\sin\theta - \sin\theta_0$, so its sidelobes are equal in that variable and the two first sidelobes of a steered beam carry the same array-factor height. The radiated pattern is the array factor multiplied by the element factor, and the element factor falls off monotonically away from broadside. The high-angle sidelobe near $+59°$ therefore sits where the element pattern is much weaker than at the low-angle sidelobe near $+8°$, and the element factor pulls it down by the difference. Asymmetry in a steered pattern is evidence of the element factor, not of a fault in the array.

(d) Real patch elements follow a power pattern nearer $\cos^{1.4}\theta$ than $\cos\theta$. Recompute the 60° scan loss and state how much worse it is than the ideal-element bound.

loss = -4.2 dB

$$10\log_{10}\left(\cos^{1.4} 60^\circ\right) = 14\log_{10}(0.500)$$
$$= -4.2 \text{ dB}$$

That is 1.2 dB worse than the -3.0 dB ideal-element bound. The $\cos\theta$ rule is a best case set by the projected aperture alone; a real element's own pattern adds to it, and a link budget that scans to 60° should carry the larger number.

4. **How far out is far enough.** The PHASER's 8 elements are spaced 14 mm apart, so the aperture spans $D = 98$ mm between the outer element centers. The HB100 runs at 10.525 GHz.

   (a) Find the far-field distance for this aperture and state whether the 1 m arc used in lab is far enough.

   $r =$ 0.67 m

   $$\lambda = \frac{3 \times 10^8}{10.525 \times 10^9} = 0.0285 \text{ m}$$
   $$r \geqslant \frac{2D^2}{\lambda} = \frac{2(0.098)^2}{0.0285} = 0.67 \text{ m}$$

   The 1 m arc is comfortably outside the 0.67 m boundary, so the pattern has settled into its far-field shape everywhere along the walk.

   (b) A Ku-band panel with a 0.200 m aperture is to be measured at 15.0 GHz. Find its far-field distance.

   $r =$ 4.00 m

   $$\lambda = \frac{3 \times 10^8}{15.0 \times 10^9} = 0.0200 \text{ m}$$
   $$r \geqslant \frac{2(0.200)^2}{0.0200} = 4.00 \text{ m}$$

   Doubling the aperture and halving the wavelength pushes the requirement out by a factor of eight.

   (c) The Ku-band panel is measured on the same 1.0 m classroom arc anyway. Find the phase error across the aperture edge and compare it against the 22.5° tolerance behind the far-field criterion.

   $\Delta\phi =$ 90°

   The extra path to the aperture edge relative to its center is $\Delta \approx D^2/8r$, and the phase error is $360^\circ \Delta/\lambda$:

   $$\Delta = \frac{(0.200)^2}{8(1.0)} = 0.005 \text{ m}$$
   $$\Delta\phi = \frac{360^\circ(0.005)}{0.0200} = 90^\circ$$

   That is four times the 22.5° the criterion allows.

   (d) Describe what a 90° edge phase error does to the measured pattern, and say which pattern feature it corrupts worst.

   The edges of the aperture arrive out of phase with the center, so the measurement no longer sees a plane wave and the aperture is effectively defocused. The main lobe broadens and its peak drops, but the damage is worst in the nulls and the sidelobes: the cancellation that produces a deep null depends on precise phase alignment across the aperture, so the nulls fill in and the sidelobes read several decibels high. A sidelobe level measured this way is pessimistic, and a null depth measured this way is meaningless.

5. **Classroom against chamber.** The lab trace carries about ± 1 dB of ripple from room reflections adding to the direct path.

   (a) List which of the four hand-measurement artifacts from the lab an anechoic chamber removes, and which one it does not remove by itself.

   A chamber removes the reflection artifact directly: absorber on every surface leaves only the direct path, so the ripple disappears. A chamber's fixtures also remove the angle error and the amplitude instability, because the antenna sits on an encoded turntable and the source is bolted at a fixed range rather than carried. The far-field condition is not removed by the absorber at all; it is a geometric requirement on the separation distance, and a chamber satisfies it only if the chamber is long enough for the aperture and frequency being measured. A large antenna in a short chamber has the same problem as a large antenna in a classroom.

   (b) A single reflected path of relative amplitude ρ adds to a direct path of amplitude 1. Find the ρ that produces a $+1$ dB peak in the received amplitude, and express it in dB.

   $\rho =$ $0.122 = -18.3$ dB

   The two paths add in phase at the ripple peak:

   $$20\log_{10}(1+\rho) = 1.0 \text{ dB}$$
   $$1+\rho = 10^{0.05} = 1.122$$
   $$\rho = 0.122 \rightarrow -18.3 \text{ dB}$$

   A reflection only 18 dB below the direct path is enough to move the reading by a decibel, which is why room reflections matter at levels that sound negligible.

   (c) The main-lobe peak and the sidelobe peak each carry their own ± 1 dB of ripple. Give the worst-case uncertainty on a sidelobe level reported in dBc, and give the range a true -13 dBc sidelobe could read.

   uncertainty = ± 2 dB

   range = -11 to -15 dBc

   A dBc value is a difference of two readings, and in the worst case the two errors have opposite signs and add:

   $$\pm 1 \text{ dB} + \pm 1 \text{ dB} = \pm 2 \text{ dB}$$

   A true -13 dBc sidelobe can therefore be reported anywhere from -11 to -15 dBc without anything being wrong with the array.

   (d) A customer requires written proof that this array's sidelobes are below -25 dBc. State whether the classroom measurement can support that claim, and give the two reasons that decide it.

   It cannot. The first reason is the uncertainty computed above: a ± 2 dB window on every dBc reading cannot certify a level that must sit below a specific number, because a measured -26 dBc could be a true -24 dBc. The second reason is dynamic range. The electrical sweep's noise floor sits about 23 dB below the peak, so a -25 dBc sidelobe is buried in the floor and there is nothing to read. Both problems are what a chamber and a calibrated receiver are for, and the claim needs a range measurement.

Documentation: _____